

Ahmad Mujtaba

Applied AI Engineer | GenAI Engineer | RAG Systems | Document AI

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Summary

Applied AI Engineer with nearly 4 years building production document AI, RAG, LLM extraction, and automation systems across healthcare payer and FMCG workflows. Built a Milvus-backed retrieval layer and Playwright MCP tooling that improved task completion from 38% to 80% on a 200-task evaluation set and reduced browser-agent prompt-token consumption by approximately 40%. Delivered multimodal prior-authorization intake, OCR/model benchmarking, and confidence-aware extraction pipelines that improved extraction accuracy from 80–81% to 90%+ and policy-entity accuracy from 90% to 99%. Strong in Python, SQL, FastAPI, Azure Databricks, Azure OpenAI, Gemini, LangChain, LangGraph, MCP-style tooling, and AI evaluation workflows.

Professional Experience

Deloitte USI | Jul 2025 – Present

AI and Data Science Engineer | Gurugram, India

- Built a multimodal document-AI and fax-intake platform for health-insurance prior-authorization workflows at a Fortune-50 payer, using Azure Content Understanding, layout-aware parsing, LLM extraction, and evaluation workflows across recurring 100-file bulk test runs.
- Engineered a Milvus-backed RAG layer for multi-agent reasoning, improving Azure Content Understanding task completion from 38% to 80% on a 200-task evaluation dataset.
- Reduced browser-agent prompt-token consumption by approximately 40% by building a Playwright MCP tool that replaced raw DOM context with accessibility-tree snapshots and JPEG-compressed observations.
- Implemented an Azure Databricks pipeline that classified fax documents by authorization type and urgency, routed only New Auth cases into extraction, and generated RPA-ready CSVs, JSON outputs, and annotated PDFs with bounding-box visualizations.
- Improved structured extraction accuracy from 80–81% to 90%+ by diagnosing OCR, layout parsing, and model reasoning bottlenecks; comparing Azure Content Understanding markdown-to-LLM flows with direct PDF/base64-to-LLM flows across GPT-4.1, GPT-5.2, Gemini 2.5 Pro, and Gemini 3 Pro; and productionizing a confidence-aware 4-pass extraction architecture with low-confidence retries, token capping, critical-field routing, and checkbox-specific vision handling.
- Raised policy-entity extraction accuracy from 90% to 99% by migrating to Gemini 3 Pro and refining canonical-comparison validation logic across 10+ prompt iterations.
- Built an automated policy eligibility assessment pipeline that used multimodal LLMs to compare scanned fax documents with policy criteria across file formats, classifying medical eligibility conditions as met or unmet while reducing manual review effort.
- Benchmarked DeepSeek-OCR, GLM-OCR (via Ollama), and PaddleOCR-VL-1.5 against cloud OCR baselines to evaluate privacy-preserving, on-prem document processing options for healthcare workloads.

Cognizant Technology Solutions | Sep 2022 – May 2025

Associate – Data Scientist / AI Developer | Noida, India

- Deployed a Spark MLlib Random Forest classifier on Azure Databricks for automotive warranty adjudication, improving recall from 79% to 88% and reducing manual review across the claims-processing pipeline.
- Engineered a conversational B2B reordering agent for a global FMCG client using AWS Lex and Azure OpenAI, automating dealer reorder workflows through AWS Lambda, Amazon S3, and structured extraction from free-text utterances.
- Migrated legacy LUIS applications to Azure Conversational Language Understanding (CLU), resolving post-migration intent/entity conflicts and improving model accuracy and recall.
- Re-architected Python applications on Azure App Services, eliminating timeout failures, improving reliability, and delivering client-requested production features.
- Built Power BI dashboards for model drift monitoring, warranty analytics, KPI tracking, and SLA compliance, enabling near real-time operational visibility and reducing executive reporting turnaround time.

Projects

Computer-Use | Open Source

React / *FastAPI* / *Docker* / *Playwright MCP* / *OpenAI* / *Claude* / *Gemini* | github.com/pypi-ahmad/computer-use

- Built an open-source multi-provider computer-use agent workbench for browser and desktop automation, running OpenAI, Claude, and Gemini actions in an isolated Ubuntu/XFCE sandbox with React, FastAPI, Docker, and Playwright MCP.
- Implemented two-phase web-search planning, action-fingerprint stuck-agent detection, async operator approvals, and real-time WebSocket screenshot/log streaming to make autonomous execution observable and controllable.

Technical Skills

Languages & Backend: Python, SQL, FastAPI, REST APIs, async services, microservices

Applied GenAI & RAG: Azure OpenAI, OpenAI (GPT-5.2/4.1), Google Gemini 3, Anthropic Claude (Opus/Sonnet), LangChain, LangGraph, Model Context Protocol (MCP), Playwright MCP, Milvus, reranking, structured outputs (Pydantic), function calling, evaluation pipelines, token-cost optimization

Document AI / OCR: Azure Content Understanding, PaddleOCR-VL-1.5, DeepSeek-OCR, GLM-OCR, PaddleOCR, layout-aware parsing, table extraction, multimodal document understanding

Data & ML: PySpark, Spark MLlib, Azure Databricks, ETL pipelines, Medallion architecture, scikit-learn, XGBoost, feature engineering, model monitoring

Local Inference & Tooling: Ollama, GGUF, quantization, browser automation

Cloud & DevOps: Azure (Databricks, App Services, Cognitive Services, Key Vault, Blob, ML Studio), AWS (Lambda, S3, SageMaker, CloudWatch, Lex), Docker, GitLab/GitHub Actions, CI/CD, Power BI

Domains: Healthcare, Prior Authorization, Health Insurance, Automotive Warranty Analytics, FMCG B2B, Conversational AI

Education

Indian Institute of Information Technology (IIIT) | Oct 2020 – Jun 2022

M.Tech, Data Analytics and Decision Sciences | Kurnool, India

- Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Statistics

Maulana Azad National Urdu University | Aug 2015 – Jun 2019

B.Tech, Computer Science Engineering | Hyderabad, India

Certifications

Machine Learning Specialization – DeepLearning.AI / Coursera | Jan 2023